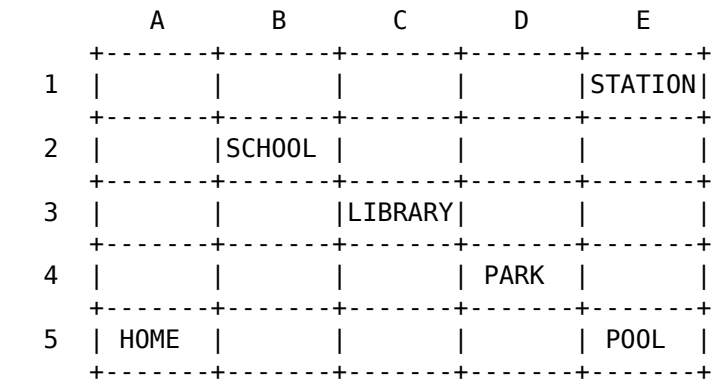
# 6.8 Shapes, maps and graphs: practice

← Unit overview · Answers

**For you:** you can use blocks, scissors, paper and a ruler for any of these. Building and folding are a great way to check your answer. No clock.

**The map for Sets 1 and 2:**




---

## Set 1 · Warm-up

1. How many sides does a triangle have? _____
2. How many faces (flat sides) does a cube have? _____
3. Which square is the school in? _____
4. What is in square C3? _____
5. Which square is the station in? _____
6. Which square is home in? _____
7. What is in square E5? _____
8. How many squares is it from home, going straight along to the pool? _____
9. How many sides does a square have? _____
10. What is in square D4? _____

---

## Set 2 · Building: scale (*schaal*)

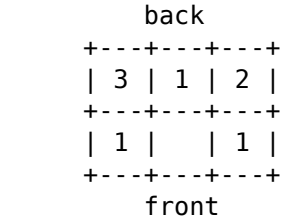
1. On a map, 1 cm is 1 km in real life. Two villages are 3 cm apart on the map. How far apart are they really? _____
2. Same map: two towns are 7 cm apart. Real distance? _____
3. On a walking map, 1 cm is 100 m. A path is 5 cm long on the map. How long is the real path? _____
4. On a cycling map, 1 cm is 2 km. A route is 4 cm on the map. How far is the real route? _____
5. Same cycling map: a real distance is 10 km. How long is it on the map? _____
6. On **our map** above, each square is 500 m wide. How far is it from home straight along to the pool? _____
7. On our map, how far is it from the station straight down to the pool? _____
8. You walk from home up to square A2 and then one square right to school. How many squares is that? How far is it? _____
9. On a floor plan, 1 cm is 1 m in real life. A table is 2 cm long on the plan. How long is the real table? _____
10. 1 cm on the map is 1 km. 1 cm on the map is _____ in real life.

**Talk about it:** In question 8 you had to walk around. Why is that longer than a straight line?

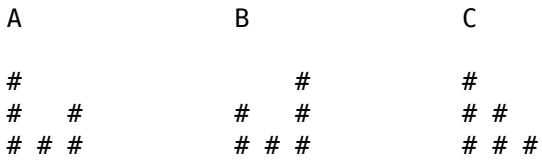
---

## Set 3 · Building: views (*aanzichten*) and nets (*bouwplaten*)

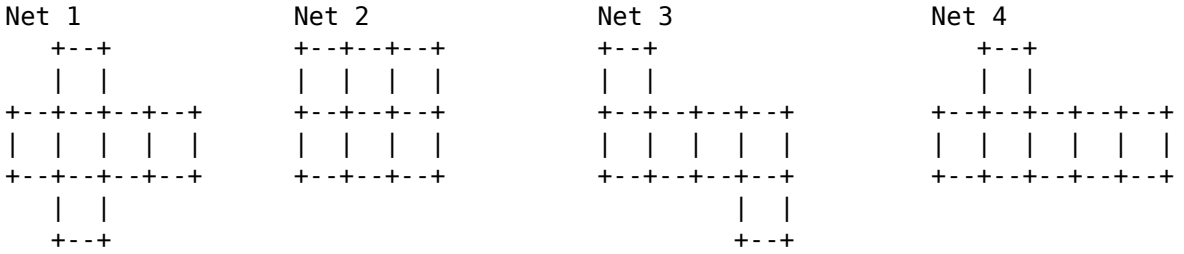
**A block building, seen from the top.** The number in each square tells how many blocks are stacked there.




1. How many blocks are there altogether? _____
2. How high is the tallest tower? _____
3. Which drawing is the **front view**? _____




**Which of these nets fold into a cube?** Write yes or no. (Cut them out from squared paper to check!)

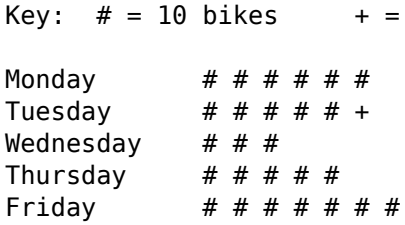



4. Net 1: _____
5. Net 2: _____
6. Net 3: _____
7. Net 4: _____
8. How many squares does every cube net have? _____

---

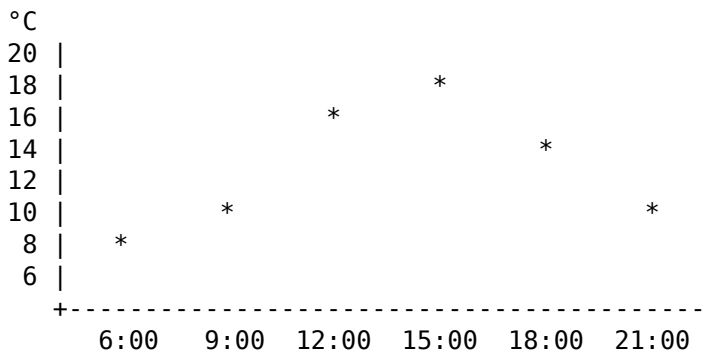
## Set 4 · Building: graphs (*diagrammen*)

**Pictogram (*beelddiagram*): bikes in the school bike shed**



1. How many bikes were there on Monday? _____
2. How many on Tuesday? _____
3. On which day were there the fewest bikes? Can you think of a reason? _____
4. How many more bikes were there on Friday than on Wednesday? _____
5. How many bikes were there altogether that week? _____

**Line graph (*lijndiagram*): temperature during one day**




6. What was the temperature at 12:00? _____
7. At what time was it warmest? _____
8. How many degrees warmer was it at 15:00 than at 6:00? _____
9. At what time was it 14 °C? _____
10. Between which two times did the temperature rise the most? _____

**Talk about it:** A shop sold 52 ice creams on Saturday and 55 on Sunday. Its bar chart starts at 50 instead of 0, so Sunday's bar looks more than twice as tall as Saturday's. Is that fair? Why might the shop do it?

_____

## Set 5 · Stretch: the ratio table (*verhoudingstabel*)

Fill in the empty boxes.

1. Drinks come in packs of 6.

| packs | 1 | 2 | 4 | 8 |
|---|---|---|---|---|
| drinks | 6 | | | |

2. A pancake recipe for 4 people needs 250 g of flour.

| people | 4 | 8 | 2 | 12 |
|---|---|---|---|---|
| flour (g) | 250 | | | |

3. A child's train ticket costs €8.

| tickets | 1 | 3 | 5 | 10 |
|---|---|---|---|---|
| euros | 8 | | | |

4. On a cycling map, 1 cm is 5 km.

| map (cm) | 1 | 3 | 7 |
|---|---|---|---|
| real (km) | 5 | | |

5. 3 apples cost €1,20.

| apples | 3 | 6 | 1 |
|---|---|---|---|
| euros | 1,20 | | |

6. *Verhaaltjessom:* You cycle 15 km in one hour. How far do you cycle in 2 hours? And in half an hour? Make a ratio table to help. _____

7. Fill in:

| 1 | 2 | 3 |
|---|---|---|
| 10 | | |

**Talk about it:** In question 2, how did you get from 4 people to 2 people? And from 4 to 12?

---

## Puzzle corner (optional)

A cube net is 6 squares, joined edge to edge, that fold up into a cube. Nets that are just turned around or flipped over count as the same net.

- Find **one** cube net that isn't on this page.
- Find **three** different ones.
- *Big challenge:* how many different cube nets are there altogether? How can you be sure you've found them all?

Squared paper and scissors are your best tools here. This one can last a whole week.